

Business Requirements Document (BRD)

Inventory Management Improvement System

Project	Inventory Management Improvement
Document	Business Requirements Document (BRD)
Prepared By	Business Analyst
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Status	Draft

1. Business Problem

The company is experiencing inventory imbalances. High-demand products frequently go out of stock, while some products remain in inventory for several months.

The current inventory process relies heavily on manual reporting and employee experience rather than structured analysis of historical sales and inventory data. This can result in delayed reordering, excess inventory, increased inventory costs, and potential lost sales.

2. Business Objective

- Reduce stockouts.
- Reduce slow-moving and excess inventory.
- Establish appropriate reorder points.
- Improve demand forecasting.
- Improve inventory visibility.
- Support data-driven replenishment decisions.

3. Stakeholders

Stakeholder	Interest
Senior Management	Revenue, profit, inventory cost, business growth
Inventory Manager	Stock levels, stockouts, excess inventory
Production Manager	Production planning and replenishment
Sales Manager	Product availability and sales
Finance Manager	Inventory cost and profitability
Wholesalers/Retailers	Product availability and order fulfillment

4. AS-IS Process

Customer Order → Sales checks inventory → If stock is available, order is processed → If stock is low, Inventory Manager checks manually → Manager decides whether to reorder based on experience → Production/Purchase is informed → Product is replenished.

Current Problems: No fixed reorder point; reordering relies on assumptions and experience; inventory reports are prepared manually; no structured demand forecasting; slow-moving products are not regularly identified; Sales and Inventory teams sometimes have inconsistent information.

5. Root Cause Analysis — 5 Whys

Problem: Popular products frequently go out of stock.

Why #1: Products are not reordered early enough.

Why #2: There is no fixed reorder point.

Why #3: Reorder decisions rely on experience and assumptions.

Why #4: Historical sales and inventory data are not analyzed systematically.

Why #5: There is no proper inventory analysis/forecasting process.

Root Cause: The company lacks a structured, data-driven inventory management and forecasting process.

6. TO-BE Process

Collect historical sales and current inventory data → Analyze demand patterns → Forecast future demand → Establish product-level reorder points → Continuously monitor inventory → If stock reaches reorder threshold, notify the responsible team → Production/Purchase decision → Inventory replenished → Monitor slow-moving products → Management receives updated inventory information.

7. Business Requirements

ID	Business Requirement	Priority
BR-01	Establish appropriate reorder points for products.	Must Have
BR-02	Use historical data for demand forecasting.	Should Have
BR-03	Base replenishment decisions on data rather than assumptions.	Must Have
BR-04	Identify and monitor slow-moving products.	Should Have
BR-05	Improve visibility into inventory levels and stock movement.	Must Have

8. Functional Requirements

FR-01 — Demand Forecasting: The system shall analyze historical sales data and generate demand forecasts for each product.

FR-02 — Inventory Monitoring: The system shall regularly monitor inventory levels and compare them against defined reorder points.

FR-03 — Reorder Alert: The system shall generate an alert when inventory reaches the defined replenishment threshold.

FR-04 — Slow-Moving Product Monitoring: The system shall identify products that meet the company's defined slow-moving inventory criteria.

FR-05 — Production Notification: The system shall notify the Production team when a product reaches its defined replenishment threshold.

9. Non-Functional Requirements

NFR-01 — Response Time: The system should notify the Production team within 1 minute after detecting that inventory has reached the defined replenishment threshold.

NFR-02 — Data Accuracy: The system should maintain accurate and consistent inventory information so that management can make reliable replenishment decisions.

10. KPIs

KPI	Purpose
Stockout Rate	Measure frequency of inventory shortages
Inventory Turnover	Measure inventory efficiency
Slow-Moving Inventory %	Monitor excess/aging inventory

KPI	Purpose
Reorder Point Compliance	Measure whether replenishment rules are followed
Order Fulfillment Rate	Measure successful customer order fulfillment

11. Expected Business Benefits

- Reduce stockouts.
- Reduce excess inventory.
- Reduce inventory holding costs.
- Improve product availability.
- Improve customer order fulfillment.
- Support better production and reordering decisions.
- Improve management visibility.
- Reduce dependency on manual judgment.

12. Next-Step BA Deliverables

- User Stories
- Acceptance Criteria
- Requirements Traceability Matrix (RTM)
- Detailed FRD